TO: ME 2171 Instrumentation Lab Instructional Staff February $24^{th}$, 2026
FROM: Mateo Jaime Tecum Jacinto, Section 07- Wednesday
SUBJECT: Lab 5, Thermocouples and Thermistors

**1.0 Statement of Purpose**

The purpose of this experiment was to analyze and observe the thermocouples voltage and thermistor's resistance to understand how their temperature measurement systems. The thermocouple circuits validate the Law of Intermediate Metals and the Law of Intermediate Temperatures.

**2.1 Background**

A thermocouple is made up of two metals that are joined together, which known as a junction. For our lab thermocouple we had two junctions, so one of these junctions was used as a reference junction. When these junctions are experiencing two different termperatures, a small voltage is produced, which is known as the Seedbeck effect. From the Seedbeck effect, the voltage created varies from both the temperature change and the material of the junctions. Using the Law of Intermediate Metals and the Law of Intermediate Temperatures. The Law of Intermediate Metals states that the addition of another metal into a circuit between two other materials will not affect the emf voltage if the other junctions are the same temperature. The Law of Intermediate Temperature states that the voltage that the thermocouple with junctions at two temperatures should be equivalent to the sum of the voltages if the thermocouple has the third temperature that is used as a reference temperature and a measured temperature. Essentially if the junction is T1 and T2 and T2 and T3 should have the same voltage as a junction with T1 and T3. The second portion of this lab involves a semiconductor device which has a resistance decrease exponentially as temperature increases. This device is known as a thermistor, and the Resistance can be calculated and represented by the following equation:

$$R = R_0 e^{\left(\beta \times \left(\frac{1}{T} - \frac{1}{T_0}\right)\right)}$$

This equation shows the following variables:

R, representing the resistance at measured temperature

$R_0$, representing the resistance at the desired reference temperature (ice bath temperature or room temp)

T, representing the measured temperature

$T_0$, The reference Temperature,

$\beta$, the material constant, which varies depending on the material

This equation shows how Resistance will change due to the material constant and the temperature, which supports the idea that these two variables determine the resistance (and as a consequence voltage as well).

## 3.0 Discussion

### 3.1 Equipment

- Copper Wire
- Constantan wire
- Iron wire
- Thermistor wire -50 kohms
- Digital multimeter
- 90 degreee temperature bath
- Keithley digital Multimeter
- Ice bath
- Hair Dryer
- Alocohol thermometer
- Banana cables with alligator clips

### 3.2 Thermocouples

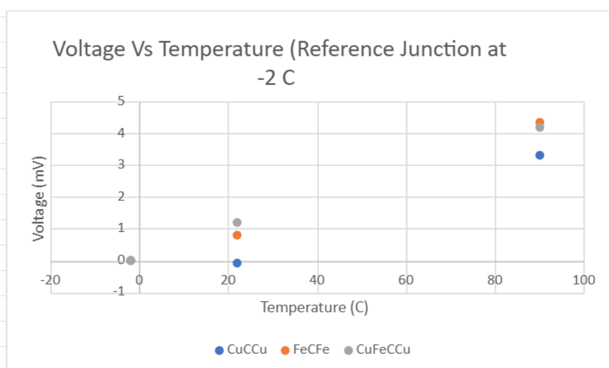


*Figure 1 Thermocouple's Voltage and Temperature graph. The reference junction was set at the temp of the ice bath*

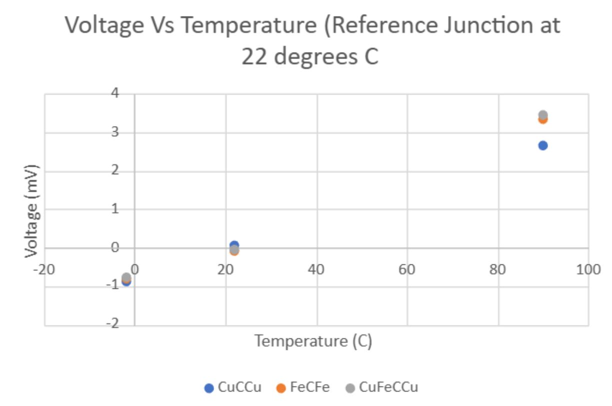


*Figure 2 Thermocouple's Voltage and Temperature relationship with the reference junction's temperature set at room temperature*

Experimenting with the thermocouples involved stripping wires to create the following junctions and circuits: CuCCu, FeCFe, CuFeCCu. For each circuit, one of the junctions was set at a reference temperature, which was either the temperature of an ice bath, or room temperature. Waiting for the circuits to cool down was performed to ensure the temperature of the junctions was the desired measured temperatures (around 5 minute wait time between each trial). For both reference temperatures, the other junction was experiencing different temperatures and the voltage was measured. The measured voltage are shown below:

| Temp | CuCCu | FeCFe | CuFeCCu |
|---|---|---|---|
| -2 | -0.011 | 0.012 | 0.01 |
| 22 | -0.0826 | 0.795 | 1.2 |
| 90 | 3.32 | 4.357 | 4.18 |
| | | | |
| | | | |
| Temp | CuCCu | FeCFe | CuFeCCu |
| -2 | -0.875 | -0.816 | -0.76 |
| 22 | 0.062 | -0.077 | -0.05 |
| 90 | 2.64 | 3.341 | 3.439 |

*Figure 3. Raw data showing the thermocouple circuits' voltage and temperature values at two different reference temperatures (-2 and 22C), voltage units were in mV*

Looking at the voltages, when the junctions were at the same temperature, the voltage measured was close to a value of zero. This indicates that the temperature had some variation(most likely due to the other junction not cooling down or warming up before taking the voltage from previous trials). Ultimately, having voltages near zero for the same junction temperatures proves that the circuit has temperatures of near equivalent values. Looking at the general equation for the circuit we see:

$$E = e_{ABT_1} - e_{ABT_2}$$

From this we can see if the voltage is close to zero we get:

$$0 = e_{ABT_1} - e_{ABT_2}$$

$$e_{ABT_1} = e_{ABT_2}$$

$$T_1 = T_2$$

Using the general equation more, an inference can be made that the voltage will increase as the temperature between T2 and T1(reference temperature) deviance increases. This is further proven from Fig 3. Which shows the 90 degree temperature having the largest voltage value.

Similarly, a different reference temperature results in the same conclusion that a bigger temperature gradient leads to higher voltages. However, since there is an ice bath temperature and a reference temperature higher than the reference, a voltage less than one is measured. This means that $e_{ABT_2}$ is greater than $e_{ABT_1}$ so the E value becomes less than 0.

Law of Intermediate Metals Sample Calculations

From the Law of Intermediate Metals we can also see that the CuFeCCu circuit is equivalent to the FeCFe thermocouple when the junctions are all at the same temperature, which is to be expected. Since both the temperatures are the same, the addition of the third material would not affect the voltage difference. The following interpretation of the Law is shown below, proving that the additional material will not affect the voltage.

$$E = e_{CuFeT_1} + e_{FeCT_2} + e_{CCuT_1}$$

$$e_{CCuT_1} = -e_{CuCT_1}$$

$$E = e_{CuFeT_1} - e_{FeCT_2} + e_{CuCT_1}$$

$$e_{CuCT_1} + e_{CuFeCT_1} + e_{FeCT_1}$$

$$E = e_{CuFeT_1} - e_{FeCT_2} - (e_{CuFeT_1} + e_{FeCT_1})$$

$$E = e_{FeCT_2} - e_{FeCT_1}$$

From this, we can see that by applying the Law to the 3rd circuit, we can simplify the junction values to cancel out the Cu values entirely, resulting in an equation that is only dependent on the Fe and C values. Fig 3. Shows similar values between the CuFeCCu and FeCFe thermocouple with the voltage values have similar values for the ice bath temp and reference temp at room temp, and the case where the reference temp at ice bath temp and the junction at ice bath temp as well. From the equation we have E represent the voltage values of the junctions temperature change. So we can also calculate the voltage from the three temperatures and verify that they are equal to each other.

Law of Intermediate Temperatures Sample Calculations

The following equation will be verified, which will validate the Law of Intermediate Temperatures:

$E_{(T_1,T_3)} = E_{(T_1,T_2)} + E_{(T_2,T_3)}$, with voltage values equal to E values of the CuCCu junction circuit

So 2.64mV=-.875mV+3.32mV

In these calculations we have $E_{(T_1,T_3)}$ represent the reference temperature of the CuCCu circuit at a reference temperature of ice bath temperature(-2 degrees Celsius), with the other junction at 90 degrees Celsius. $E_{(T_1,T_2)}$ will represent the circuit at a reference of -2 degrees Celsius(ice bath temp), with the other junction at room temperature. Finally the $E_{(T_2,T_3)}$ represents the voltage value when the reference temperature was room temperature(22 degrees Celsius) and the other junction was at a temperature of 90 degrees. From this, we can see if the law of Intermediate temperature is valid, then the voltage of the reference at -2 degrees C, with 90 degrees should be equivalent to the sum of the other two voltages.

Although the difference between 3.32mv and -0.875mV is 2.445mV, the deviation is most likely due to measurement errors, and slight temperature measurements errors. The error percentage for this turns out to be 7.97%, which is reasonable. However, since the measurements are close enough, it is still valid to state the Law of Intermediate Temperatures stands.

3.3 Thermistor

The following raw data is given for the thermistor experiments:

| Temp C | Res (k-ohms) | | | K | x | y |
|---|---|---|---|---|---|---|
| 0 | 152 | | | 273.15 | 0.00366 | 0.0000 |
| 22 | 57.57 | | | 295.15 | 0.00339 | -0.9709 |
| 32 | 34 | | | 305.15 | 0.00328 | -1.4975 |
| 65 | 6.63 | | | 338.15 | 0.00296 | -3.1323 |
| 90 | 4.73 | | | 363.15 | 0.00275 | -3.4700 |

*Figure 4. Raw data of thermistor resistance and temperature values in Celsius and kelvin, and resistance in kOhms. X and Y values are calculated for linear model.*

When switching over to the thermistor portion of the experiment, our new variables include Resistance and temperature. The thermistor itself, does not involve creating (stripping wires) circuits of different material. The equation that represents the relationship of the thermistor's resistance and temperature is the aforementioned equation,

$$R = R_0 e^{(\beta \times \left(\frac{1}{T} - \frac{1}{T_0}\right))}$$

This equation was both used and also manipulated to represent our experiments as an exponential decrease, and as a linear model. To linearize our data points the following was done to the equation:

$$R = R_0 e^{(\beta \times \left(\frac{1}{T} - \frac{1}{T_0}\right))}$$

$$\frac{R}{R_0} = e^{(\beta \times \left(\frac{1}{T} - \frac{1}{T_0}\right))}$$

$$Ln(\frac{R}{R_0}) = Ln[e^{\left(\beta \times \left(\frac{1}{T} - \frac{1}{T_0}\right)\right)}]$$

$$Ln(\frac{R}{R_0}) = \left(\beta \times \left(\frac{1}{T} - \frac{1}{T_0}\right)\right)$$

$$Ln\left(\frac{R}{R_0}\right) = \beta \frac{1}{T} - \beta \frac{1}{T_0}$$

This new equation can be plotted where the independent variable is $\frac{1}{T}$ and the dependent variable y is represented as $Ln\left(\frac{R}{R_0}\right)$. In this way, the new equation has the same form as a y=mx+b form. Therefore when plotting both graphs we will obtain an exponential graph that is decreasing as temperature increases, and a linear graph that increases as Temperature decreases( 1/T increases). The follow figures are shown below:

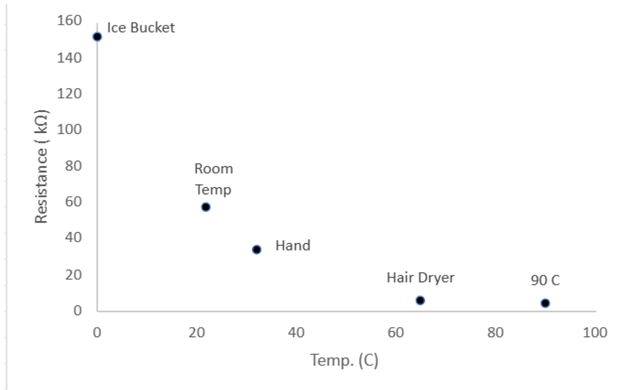


*Figure 5. Experimental thermistor graphed data showing the relationship between resistance and temperature(K) in 5 different temperature conditions, with the reference temperature being the temperature of the ice bath temperature*

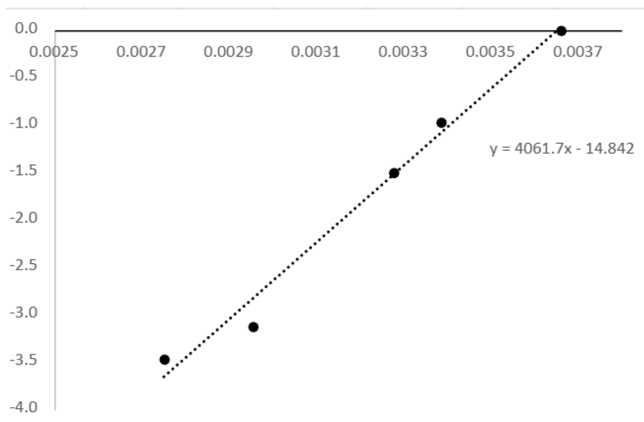


*Figure 6. Linearized thermistor graph showing a increasing linear line, with a slope representing the material constant, $\beta$ = 4061.7., reference temperature at -2 degrees Celsius*

Given the X vs Y model of the manufacturers points result in similar plots(shown below), the graphs above show the expected behavior between resistance and temperature. For the linearized model, the slope was seen as 4061.7, which varies slightly from the manufactures 3961 slope value, but this is expected as there are some uncertainties within our measurements. From this equation and our linearized model's slope, we can claim that variable, **$\boldsymbol{\beta}$ = 4061.7 K**. Additionally, our y-intercept value is very similar to the manufacturer's measurements, which means our $\beta$ is very close.

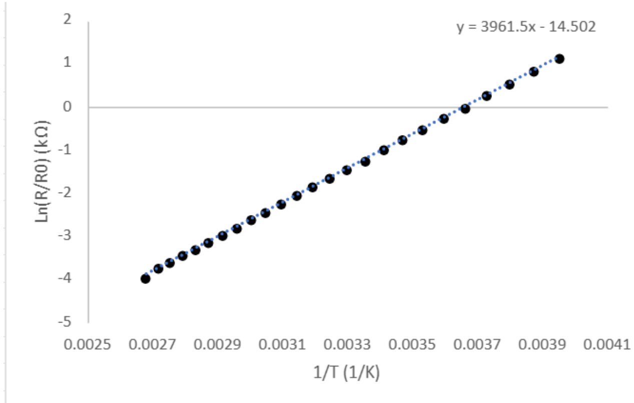


*Figure 7. Manufacter data values' plotted in the linear equation, with $\beta = 3961.5K$, reference temperature was 0 degrees celsius*

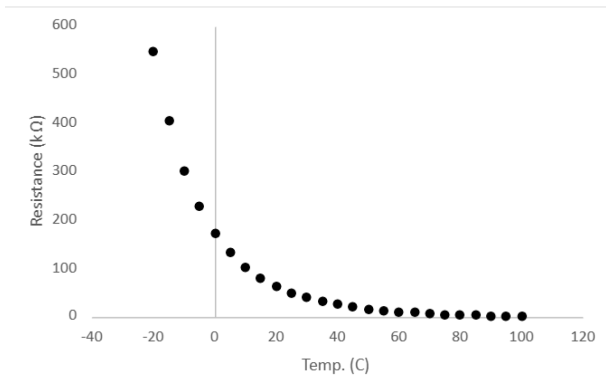


*Figure 8. Manufacturer thermistor data points plotted, showing an exponential decrease of resistance as temperature increases.*

## 4.0 Conclusions

The lab observations and measurement emphasize the Laws of Intermediate Temperature and the Law of Intermediate Metals. The Law of Intermediate metals was proven as our large circuit, CuFeCCu, produced similar values of voltages as the FeCFe when the reference temperature of the junctions were the same as the junction that had a varying temperature. Additionally, from the Thermistor experiments verified that Resistance relies on temperature, and with our data we were able to deduce that the β value equals 4061.7 K. This value does deviate from the manufacturer's calculated data slope but is within a reasonable range.